



Kian Zarazvand

I design suspension, mechanisms and cryostat hardware — then build them in house and find out what breaks.

I'm a mechanical engineering student at the University of California, Riverside, expecting to graduate in 2029. I started in FIRST Robotics in 2022 on Team 6560 Charging Champions, ran a 30-student team as captain, and now mentor two programs while working on suspension for UCR Formula SAE and cryogenic hardware for the Dark Matter & Neutrino Lab. Most of what I know came from iterating on a part until it stopped breaking.

DEGREE	BS Mechanical Engineering, UC Riverside · expected June 2029
GPA	3.83
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Experience

May 2026 – present	Mechanical Engineer Dark Matter & Neutrino Lab, UC Riverside
Sep 2026 – present	Undergraduate Researcher, simulation Concentric tube robotics research, UC Riverside
Oct 2025 – present	Suspension Intern → Associate Engineer UCR Formula SAE
Jun 2025 – present	Project Manager & Design Mentor FIRST Robotics
May 2024 – May 2025	Team Captain FIRST Robotics, Team 6560
Aug 2023 – Apr 2025	CAD Lead FIRST Robotics, Team 6560
Jun 2023 – Sep 2025	Design Engineer, elevator mechanisms FIRST Robotics, Team 6560
2022 & 2023	Assistant Stage Manager University High School Technical Theatre

Skills & tools

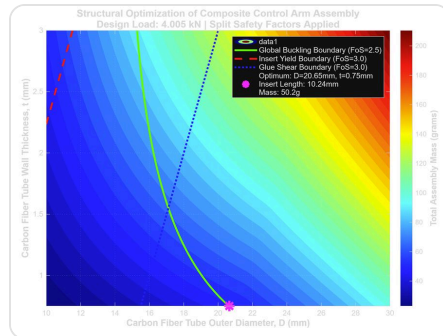
- CAD and simulation: SolidWorks (design and simulation), Fusion 360 (CAD/CAM), Onshape, FEA, skeleton modeling, GD&T, Design for Manufacturability.
- Manufacturing: CNC machining, laser cutting, 3D printing, sheet metal, carbon fiber layup, billet machining, welded and epoxy-bonded joints.
- Analysis: MATLAB optimization, multivariable calculus, linear algebra, stress calculations, simulation-testing code.
- Project management, part procurement, technical mentorship, sponsorship and budgeting.
- Power transmission: belts, #25 chain, gear racks, bevel gears, custom gearboxes, turntables.
- Sensing and control integration: hall-effect sensors, relative encoders.

Education

- BS Mechanical Engineering, University of California, Riverside — 3.83 GPA, expected June 2029.
- Coursework: MATLAB, multivariable calculus, linear algebra, GD&T, 3D CAD design.
- University High School, Irvine — graduated June 2025 with a 4.15 GPA.

Formula SAE suspension

ANALYSIS



Carbon fiber pushrod

Associate Engineer, Suspension • May 2026

A pushrod only has to carry axial load, which makes it the ideal first part to move from steel to a bonded carbon tube. I wrote a MATLAB optimizer that sweeps tube diameter and wall thickness against global buckling, insert yield and glue shear limits, then sized the insert and bond length from the result.

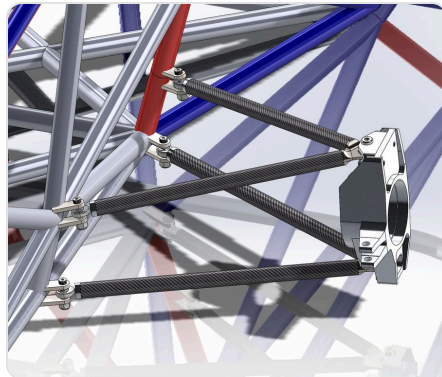
DESIGN ELEMENTS

- MATLAB optimizer for tube sizing and wall thickness — 4.005 kN design load with split safety factors (buckling FoS 2.5, insert yield FoS 3.0, glue shear FoS 3.0).
- Optimum landed at 20.65 mm outer diameter, 0.75 mm wall and a 10.24 mm insert length for a 50.2 g assembly.
- FEA-optimized aluminum insert.
- Epoxy bonding research and development for the tube-to-insert joint.

FUNCTIONALITY

- 60% lighter than the prior steel pushrod.
- Lowers unsprung mass for better vehicle dynamics.
- Proves out the carbon structure and adhesive bonding for future full carbon fiber control arms.

CAD



Carbon fiber control arm

Suspension Intern • January 2026

Suspension for a Formula Student car has to be light enough to matter and stiff enough to survive a season of testing. This intro project took the control arm from skeleton model through FEA to a preliminary carbon fiber design bonded into billet inserts.

DESIGN ELEMENTS

- Skeleton modeling, FEA simulations and stress calculations with Design for Manufacturability in mind.
- Steel tabs to attach to the chassis.
- Billet steel insert bonded into the carbon tubes.
- Epoxy tube inserts, ball joints, rod ends and welded joints for a precise, configurable yet simple design.
- Iterated on billet-machined components, sheet-metal structures and metal plates to cut weight and cost.

FUNCTIONALITY

- Preliminary carbon fiber suspension design for the car.
- Higher stiffness than the tubular steel arms it replaces.

Cryogenic research hardware

01 CAD



02 PROTOTYPE



Lumirror sheet housing

Mechanical Engineer · May 2026 – present

The housing has to keep a 1 m × 8 m reflective plastic sheet taut and located inside a cryostat while the whole assembly contracts from room temperature down to 70 K.

DESIGN ELEMENTS

- CNC aluminum for the prototype.
- Laser-cut 304 stainless steel for the final product.
- Sheet metal tabs and clamping sheets to grip the film without tearing it.
- Fastening designed around rapid cryogenic thermal contraction.

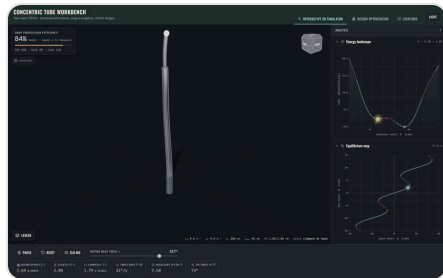
FUNCTIONALITY

- Tensions and houses a 1 m × 8 m sheet of Lumirror plastic.
- Withstands rapid thermal contraction from room temperature to 70 Kelvin.

UCR UNDERGRADUATE RESEARCH · CONCENTRIC TUBE ROBOTICS

Snap-driven swimming robot

ANALYSIS



Concentric tube snap simulator

Undergraduate Researcher, concentric tube robotics · UC Riverside · Sep 2026 – present

Medical concentric tube robots treat the snap of pre-curved tubes as a failure to avoid. Our group uses it on purpose: a fin that stores torsional energy and releases it in a burst-and-coast swimming stroke. Since joining the group I have been responsible for the simulator, using it to re-evaluate whether our original fin plan was the most efficient and to explore other ways of harnessing the Nitinol snap to propel the robot.

Live tool: qwertz101.github.io/turtle-robot-sim/

DESIGN ELEMENTS

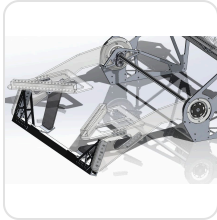
- Two-tube model built on the Gilbert–Hendrick–Webster elastic stability analysis: bifurcation parameter λ , snap onset at $\lambda_0 = \pi^2/4$, energy landscape and equilibrium map.
- Design optimization surface of physical snap energy $E_{\text{snap}} = \Delta E \cdot k_t / L_c$ (mJ) across precurvature and overlap length, scored by propulsion efficiency and Nitinol fatigue strain.
- Free, fully constrained and planar outer-tube modes to compare how the snap energy splits between bending and torsion.
- Photoreal three.js view with physically consistent shadows, slow-motion snap replay and a SolidWorks-style view cube.

FUNCTIONALITY

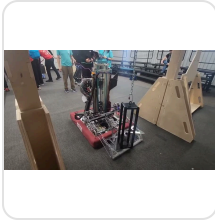
- Traces the fin's motion path, colour-coded build-up vs snap, and resolves a net propulsion vector shown as a compass.
- Snap propulsion efficiency: the share of each snap that pushes along the net propulsion direction.
- Every quantity carries units and its source equation; a citations page covers every paper used.
- Deployed on GitHub Pages so the whole group can use it from a browser.

Competition robot mechanisms

01 CAD



02 IN COMPETITION



Climbing rotary mechanism

Team Captain & CAD Lead · February 2025

The end-game climb had to catch a swinging steel cage, then rotate it into the robot hard enough to lift 150 lb off the floor — and survive being slammed into the cage at speed, match after match.

DESIGN ELEMENTS

- Utilized 3/8" polycarbonate for robustness.
- Engineered a gusset-and-tube structure for simplicity.
- Created an impact-resistant funnel with polycarbonate plates.
- Used sheet metal bending to lower assembly complexity.

FUNCTIONALITY

- Funnels a steel "cage" into the mechanism.
- Lifts the 150 lb robot by rotating the cage into the robot.

01 CAD



02 PROTOTYPE 03 BUILT



Offset pivot rotary mechanism

Team Captain & CAD Lead · January 2025

The arm needed to pivot from the top of the elevator without its gearbox riding along for the trip. Offsetting the pivot through a large turntable and a chain run let the gearbox sit low while the arm stayed stiff.

DESIGN ELEMENTS

- Used #25 chain for lightweight, high-torque power transmission.
- Simplified the design with an inline turnbuckle chain tensioner.
- Increased joint stiffness with a large-diameter turntable.
- Used an external relative encoder to account for gearbox backlash.
- Lowered the gearbox to lower the mechanism centre of gravity.

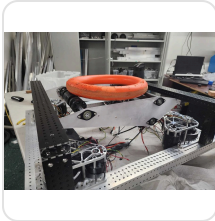
FUNCTIONALITY

- Offsets the pivot of an "arm" mechanism.
- Integrates with the "elevator" mechanism.

01 CAD



02 PROTOTYPE



Transfer mechanism

CAD Lead · July 2024

A game piece has to move from the intake to the shooter in a fraction of a second without getting pinched or thrown. The transfer is a short conveyor built from polycarbonate and belted rollers.

DESIGN ELEMENTS

- Designed a high-speed conveyor belt.
- Used polycarbonate for a lightweight structure.

FUNCTIONALITY

- Transfers game pieces from one mechanism to another.

CAD



Launch angle rotary mechanism

CAD Lead · July 2024

Instead of driving the shooter pivot through its own axis, a curved gear rack lets a small pinion hold the whole assembly at any angle with plenty of torque — and the rack doubles as structure.

DESIGN ELEMENTS

- Used a 20DP gear rack for stability and high torque.
- Reinforced the structure with carbon fiber tubes.
- Aggressive weight reduction with pocketing.
- Customized the power transmission for optimal integration.
- Precisely integrated with the surrounding subsystems.

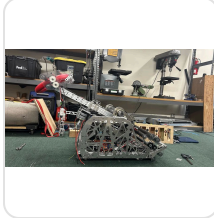
FUNCTIONALITY

- Rotates the "shooter" mechanism up and down.
- Moves game pieces from the "transfer" to the "shooter".

01 CAD



02 BUILT



2-stage linear motion + rotational gearbox

CAD Lead · January 2024

Packaging is the whole problem here: the wrist gearbox lives inside the elevator carriage, the belt is diverted around it, and a constant-force spring cancels the mechanism's own weight so the motor only moves the payload.

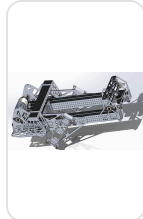
DESIGN ELEMENTS

- Used a bevel gear to optimize the gearbox packaging.
- Diverted the belt for packaging.
- Utilized a 10 lb constant-force spring to offset the mechanism weight.

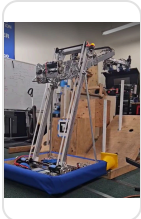
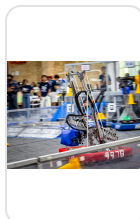
FUNCTIONALITY

- Extends a "wrist" mechanism to score game pieces.
- Houses the "wrist" gearbox in the mechanism carriage.

01 CAD



02 BUILT

03 IN
COMPETITION

3-stage linear motion mechanism

Design Engineer · July 2023

The mechanism I have iterated on longest. Cascade rigging lets one motor drive three stages, and the belt tensioning was redesigned until it was continuous, easy to make and did not skip under load. Over twelve revisions across two years, the family of elevators that grew from this design reached 60 inches of travel in 0.4 seconds and a 300 lb lifting capacity.

DESIGN ELEMENTS

- Utilized cascade rigging to extend multiple stages with one motor.
- Customized the belt tensioning to be continuous and easily manufacturable.
- Pocketed CNC plates to meet weight limits.
- Iterated over 7 times to perfect the design.
- Later revisions added A-frames, custom gearboxes and hall-effect sensors for reliability and control.

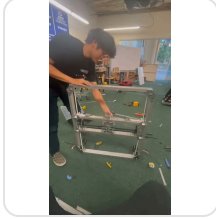
FUNCTIONALITY

- Moves items double the length of the mechanism.
- Transports game pieces to score at a distant location.
- Design family rated for 60 in of travel in 0.4 s and 300 lb lifting capacity.

01 CAD



02 BUILT



3D printer training task

New member training project • June 2023

A two-axis gantry in the style of a 3D printer, built as a training exercise that became the team's testbed for belted linear motion.

DESIGN ELEMENTS

- Protected the powertrain by belting inside the tubestock.
- Tensioned the belt to minimize backlash.
- Fortified the structure with corner gussets.
- Used 6061 aluminum tubestock for the structure.

FUNCTIONALITY

- Provides research and development for future linear motion mechanisms.
- X and Y axis degrees of freedom.

One mechanism, re-optimized every season

I have been the team’s elevator specialist since 2023. Each season the game asked for something different — more reach, less weight, tighter packaging, more load — so the same linear-motion mechanism was re-optimized around new requirements. Over 12 revisions in two years, the family reached 60 in of travel in 0.4 s and a 300 lb lifting capacity.

	MULTI-STAGE	LIGHTWEIGHT	COMPACT	SIMPLE	HEAVY-DUTY	FAST
V0 Gantry trainer Jun 2023				●		
V1 3-stage cascade elevator Jul 2023	●				●	
V2 2-stage elevator + rotational gearbox Jan 2024			●	●		
V3 Elevator with offset arm pivot Jan 2025					●	●

BUILT



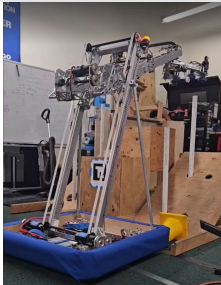
V0 · Jun 2023

Gantry trainer

Requirement: Learn belted linear motion before building it on a robot.

- Belts run inside 6061 tubestock to protect the powertrain.
- Belt tensioned to minimize backlash; corner gussets for stiffness.
- Became the testbed for every elevator after it.

BUILT



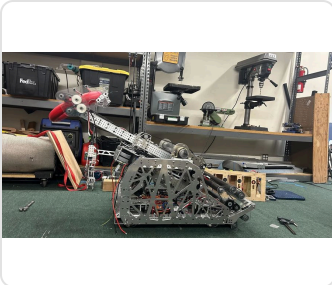
V1 · Jul 2023

3-stage cascade elevator

Requirement: Score at a distant location: travel double the mechanism length.

- Cascade rigging drives three stages from one motor.
- Continuous, easily manufactured belt tensioning.
- Pocketed CNC plates to meet the weight limit; 7+ iterations.

BUILT



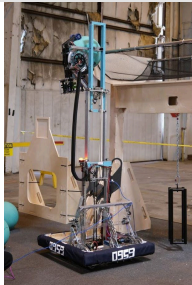
V2 · Jan 2024

2-stage elevator + rotational gearbox

Requirement: Carry a powered wrist without growing the robot.

- Dropped to two stages; the wrist gearbox lives inside the carriage.
- Bevel gear and a diverted belt path for packaging.
- 10 lb constant-force spring cancels the mechanism weight.

BUILT



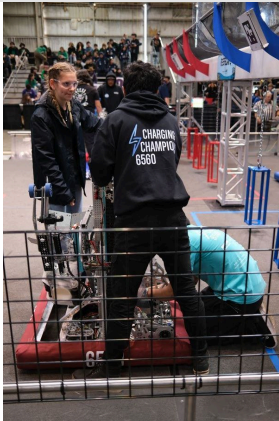
V3 · Jan 2025

Elevator with offset arm pivot

Requirement: Hold a long pivoting arm at full height without flex.

- Large-diameter turntable stiffens the arm joint.
- #25 chain with an inline turnbuckle tensioner.
- Gearbox lowered to drop the mechanism centre of gravity.

Running the team



Team captain

Team Captain, Team 6560 • Irvine, CA • May 2024 – May 2025

Captaining meant owning the whole pipeline — not just the robot, but the people, money and schedule that produce one on a six-week clock.

DESIGN ELEMENTS

- Led a 30-student team in designing, building, wiring and coding a competition robot in two months.
- Managed recruitment, reimbursements, sponsorships and robot integration across subteams.
- Handled project management across the mechanical, electrical and software groups.

FUNCTIONALITY

- Competed in five yearly competitions with the resulting machine.



CAD lead

CAD Lead, Team 6560 • Irvine, CA • August 2023 – April 2025

This was where the team's design practice got built: a shared library, a file system people could actually navigate, and six designers trained to the same standard.

DESIGN ELEMENTS

- Managed a team of 6 designing 400+ component robots with CNC, laser cutter and 3D printer in-house manufacturing and prototyping.
- Led R&D efforts to expand the design library and organized the team hierarchy for better productivity across skill levels.
- Created a Google Drive file management system and managed part procurement.

FUNCTIONALITY

- Trained new members to SolidWorks proficiency.
- Two full competition robots shipped on schedule.



Design mentor & project manager

Irvine High School Robotics & Charging Champions Robotics • June 2025 – present

Two strands of the same skill: getting a group of people to deliver a working physical system on a fixed date.

DESIGN ELEMENTS

- Provide design mentorship and CAD training for Irvine High School Robotics and Charging Champions Robotics.
- Project management for a controls team working in automation, vision, object detection and general robot programming.
- Implemented concurrent design and manufacturing, and standardized simulation-testing code.

FUNCTIONALITY

- As Assistant Stage Manager for University High School Technical Theatre, managed a crew of 10 through the technical set-up and planning of lighting, scenery and audio for productions (2022, 2023).